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Palaeomagnetics and igneous activity of the area adjoining the south-western margin of the Cuddapah basin, India

Anil Kumar and M. S. Bhalla *Palaeomagnetics Group, National Geophysical Research Institute, Uppal Road, Hyderabad 500 007 (A.P.), India*

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Summary. Fifty-six orientated samples were collected from 13 sites on five dolerite dykes (between lat. 14°23′N, long. 77°43′E and lat. 14°08′N, long. 77°49′E), which adjoin the south-western margin of the Cuddapah basin in Anantapur district, Andhra Pradesh. After af demagnetization, two dykes (five sites) striking ENE possess similar magnetic directions, (1) $D = 57°$, $I = -69°$ ($K = 52$, $\alpha_{95} = 7°$) and (2) $D = 71°$, $I = -72°$ ($K = 260$, $\alpha_{95} = 5°$). Again dykes (3) (three sites), and (4) (two sites) have similar strike (NE) and magnetic directions, $D = 64°$, $I = -7°$ ($K = 142$, $\alpha_{95} = 8°$) and $D = 53°$, $I = -8°$ ($K = 142$, $\alpha_{95} = 6°$) and dyke (5) (two sites) striking NW shows $D = 320°$, $I = -34°$ ($K = 68$, $\alpha_{95} = 13°$). Remanent directions estimated from total field magnetic anomaly data agree well with these results. Synthesis of these data with 10 other published palaeomagnetic studies of Precambrian dolerite dykes on the Indian peninsula, suggest that these three systems of dykes adjoining the Cuddapah basin had been emplaced prior to the basin formation perhaps representing the initial thermal event responsible for the basin development and also that there have been at least three separate periods of dyke emplacement on this shield. The radiometric data, however, are very sparse and these periods cannot be dated with confidence.

Introduction

Among the world's Precambrian basins, the Cuddapah basin (an intracratonic basin) in the Indian peninsular shield is possibly one of the best studied. Geophysical studies which include gravity, magnetic (Balakrishna & Paul 1970; Balakrishna 1979; Rao, Rao & Raju 1979), deep-seismic sounding (Kaila *et al.* 1979), electrical resistivity (Balakrishna 1979) and telluric (Sanker Narayan *et al.* 1979) studies etc., carried out in and around the intracratonic basin have revealed the deep hidden structures (mantle perturbations), basin configuration, depth of the sedimentary pile, underlying crust–mantle structures and the influence of the deep-seated process in the development of the basin (Bhattacharji 1981).

Magnetic studies across dykes and sills in the south-western section of the lower Cuddapahs (Vempalli & Tadpatri formations, Table 1), and the area surrounding the basin (Balakrishna 1979) reveal different intensity of magnetization in the dykes (in the basin and

Table 1. Stratigraphic column of the Cuddapah basin sequence (after Murthy 1979b).

Kurnool group		
	Unconformity	
Srisailam quartzite		Igalapenta quartzite Tapasipenta siltstone Kaklet vagu quartzite
	Unconformity	
Nallamalai group		Cumbum formation Shale/phyllite, quartzite/dolomite Bairenkonda quartzite
	Angular unconformity	
Chitravati group		Gundikota quartzite Tadpatri formation (including sills etc.) Pulivendla quartzite
	Disconformity	
Papaghni group		Vempalli formation (dolomite, shale, lava flows etc.) Gulcheru quartzites
	Non-conformity	
Archaean granites, gneisses, schists		

its surrounding) indicating that the igneous activity of these two regions had taken place at different times (Balakrishna, Paul & Ramana Rao 1972). Chemically also these two basaltic suites have varying composition (Bhattacharji 1981). This apart, magnetic data cross dykes (nine in all) lying to the south-west of the basin were interpreted in terms of their magnetic directions (vector sum of remanent and induced components, Balakrishna 1979) with the aid of Heiland's (1968) equations. Subsequently these nine dykes were grouped into two – one almost horizontally magnetized and the other vertically. These results are in agreement with the palaeomagnetics presented here.

Intracratonic basins (like the Cuddapah basin) are a structure superposed on the continental crust. Their inception and development are closely controlled by the condition of the crustal stability and its isostatic imbalance. Isostatic imbalance may be brought about by an initial thermal event immediately before the start of subsidence (Sleep 1971; Sleep & Snell 1976). The igneous activity (dykes and sills in the lower Cuddapahs) in the basin is younger than the basin itself (Balakrishna 1979 and Murthy 1979a). Therefore, the initial thermal event which preceded the basin formation may be represented by the large dyke swarm surrounding the basin. Palaeomagnetic studies of dykes from this swarm may be helpful in establishing their relationship with the adjoining intracratonic (Cuddapah) basin.

Indian Proterozoic apparent polar wandering path is not sufficiently detailed. Therefore, synthesis of published palaeomagnetic results of Precambrian dykes from various parts of the Indian Peninsular Shield with similar remanent magnetic directions (RMD) to those of the present study was undertaken in order to make a comparative study and attempt to assign plausible ages to these dykes. Incidentally this study could also furnish better knowledge about the dyke activity during the Precambrian times on this shield.

Geology and sampling

The Indian Peninsular Shield is infested by a large number of igneous dykes which constitute swarms in a few localities. One such dyke swarm is found adjacent to the south-western margin of the Cuddapah basin (Fig. 1). The basement rocks of this area are peninsular gneisses and schists (Raju, Mohd & Prabhakar Rao 1979). These dykes, which may be of different ages, strike in different directions, NE–SW, ENE–WSW or NW–SE, and are seen as persistent linear ridges extending for several kilometres with widths varying between 5 and 20 m. Some of the dykes run towards the Cuddapah basin and abruptly end at its margin (e.g. Fig. 1, dyke ii) perhaps continuing under the sediments without cutting them, in which case such dykes could be older than the basin. Gravity and magnetic studies carried out in the area have revealed the existence of dykes in the basement rocks below the Cuddapah sediments. These dykes occupy an area of about 3000 km^2 in the south-western region of the basin (Balakrishna, Paul & Ramana Rao 1970; Balakrishna & Paul 1970).

Petrographic studies have been carrried out on many of the samples. They are medium-grained with plagioclase, augite, quartz, magnetite and hypersthene as major constituents. Poikilitic texture is very common though in a few sections porphyritic texture is also observed. Fresh magnetite grains with two sets of exsolved ilmenite lamellae were identified with an ore microscope. The modal abundance of magnetite is 4–5 per cent.

Almost all the samples for this study were collected from fresh exposures in streams and rivulet beds: this being the only reason why there is an uneven distribution of the sites sampled (Fig. 2). In all, five dykes have been sampled. Of these, i and ii run ENE–WSW, iii and iv NE–SW; and v NW–SE, which means that at least one dyke in the three principal strike directions of the swarm has been sampled. As shown in Fig. 2, dykes ii and iii (near Bukkapatnam) intersect each other. Careful observation at the intersection of these two dykes shows that dyke iii has been cut by dyke ii, which is therefore, the younger one. Three

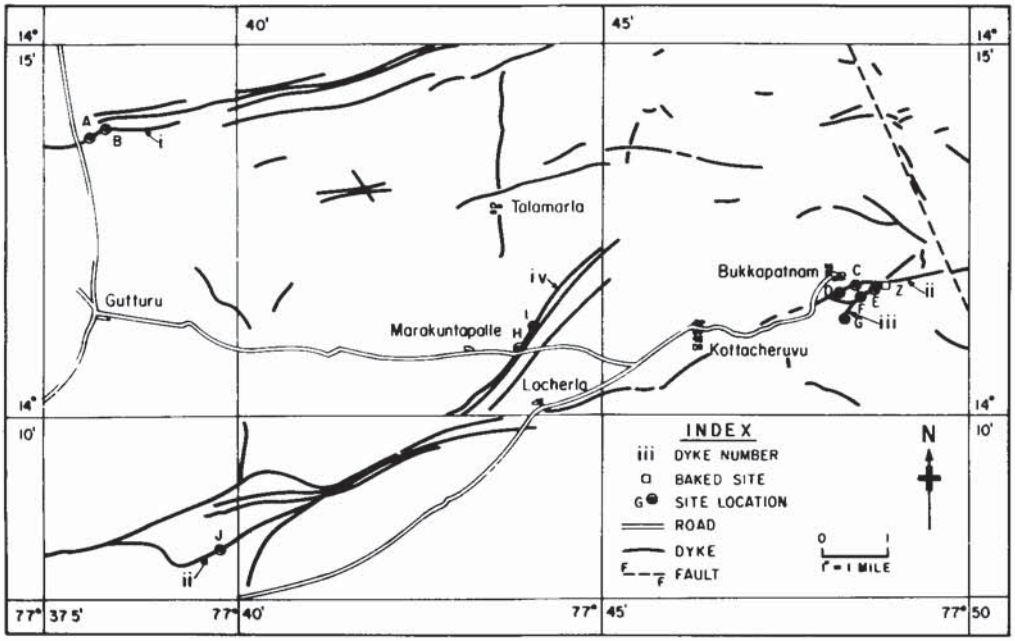


Figure 2. Map showing all the dykes in the area, sampling sites are indicated by symbols marked A–J and dykes by (i)–(iv).

Table 2. Results of palaeomagnetic measurements.

Dyke	Site	N	n	NRM					Treatment	After af demag.				Pole position	
				D_m	I_m	$J \times 10^{-7}$ Am^2	K	α_{95}	mT	D_m	I_m	K	α_{95}	λ_P	L_p
i	A	5	19	44	−33	9.6	5	35	40	56	−68	70	6	8 N	225 E
	B	5	17	58	−45	2.1	17	15	40	57	−71	36	8	6 N	229 E
ii	C	4	16	47	−59	4.2	141	6	40	88	−73	185	3	11 S	226 E
	D	4	15	47	−49	3.5	13	20	20	71	−74	547	2	4 S	229 E
	J	5	10	334	+66	5.0	9	23	40	59	−70	48	10	5 N	226 E
iii	*Z	3	12	28	−62	7.8	11	13	40	56	−68	55	11	9 N	227 E
	E	5	16	61	+22	16.2	13	19	20	60	−08	76	7	26 N	179 E
	F	3	11	59	+8	17.6	82	6	20	66	−05	104	6	21 N	176 E
	G	4	13	68	−9	30.9	106	8	20	65	−09	247	11	21 N	178 E
iv	H	4	15	33	+11	14.2	44	11	20	55	−05	95	5	33 N	181 E
	I	5	16	46	+24	10.1	22	15	20	59	−11	75	6	29 N	182 E
v	K	5	14	348	+33	2.4	37	12	15	321	−37	18	19	39 S	126 E
	L	4	14	346	+25	3.5	41	13	15	314	−27	118	8	37 S	137 E

N = number of samples; n = number of specimens; D_m = mean declination; I_m = mean inclination; J = intensity; α_{95} = radius of circle of confidence at 95 per cent probability; K = precision parameter; λ_P and L_p = palaeocoordinates of the pole; *Baked site; statistical parameters based on sample means.

samples were deliberately collected on dyke iii very near the point of intersection, to study their magnetic direction. Rest of the samples from these two dykes were collected 30–35 m away from the actual intersection. Three to five oriented block samples were collected from each of the 13 sites, two sites each on dykes i, iv and v, three on dyke ii, and four on dyke iii.

Laboratory measurements and analysis of data

The oriented block samples were cored and cut into cylinders of 2.5 cm length and 2.5 cm diameter with a minimum of three cylindrical specimens being obtained from each block sample, resulting in 175 specimens in all.

Remanent magnetism was measured using an astatic magnetometer. The NRM directions which do not show much scatter are given in Table 2. Two specimens from each site were progressively demagnetized in ten steps up to a peak af value of 80 mT. The migrating magnetic vectors and intensity decay curves during progressive af demagnetization are represented in Figs 3 and 4. On the basis of these results, the optimum field for batch demagnetization of all samples was selected. At least two specimens from each sample were subjected

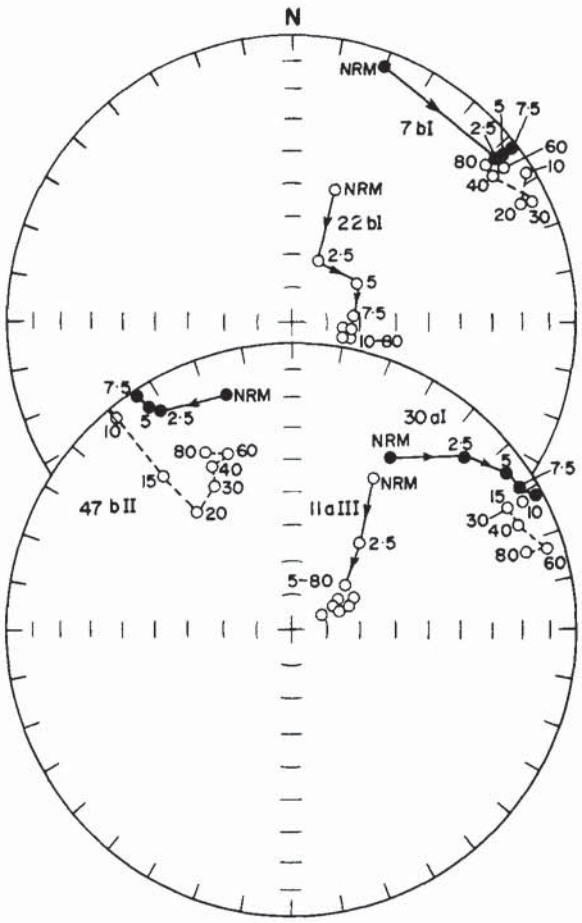


Figure 3. Equal area projection showing directional changes during af demagnetization peak fields in mT. Solid circles indicate positive dips, open circles negative dips. Sample 7bI represents site A, dyke i; 11aII represents site C, dyke ii; 30aI represents site H, dyke iv and 47bII represents site K, dyke v.

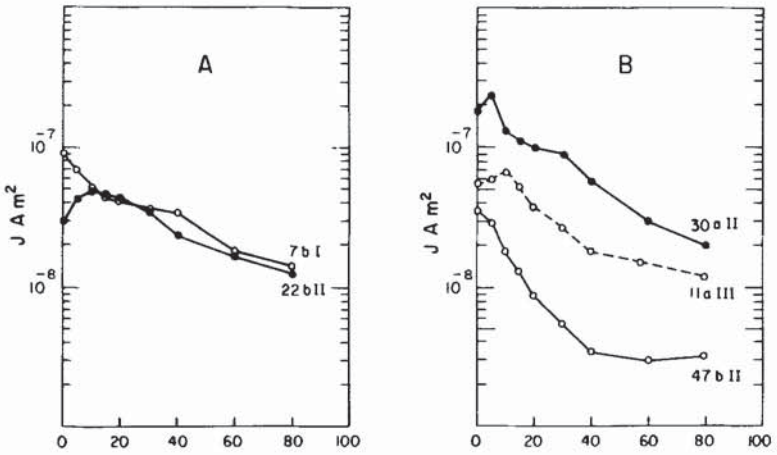


Figure 4. Decay of remanent intensity on af demagnetization, peak field in mT.

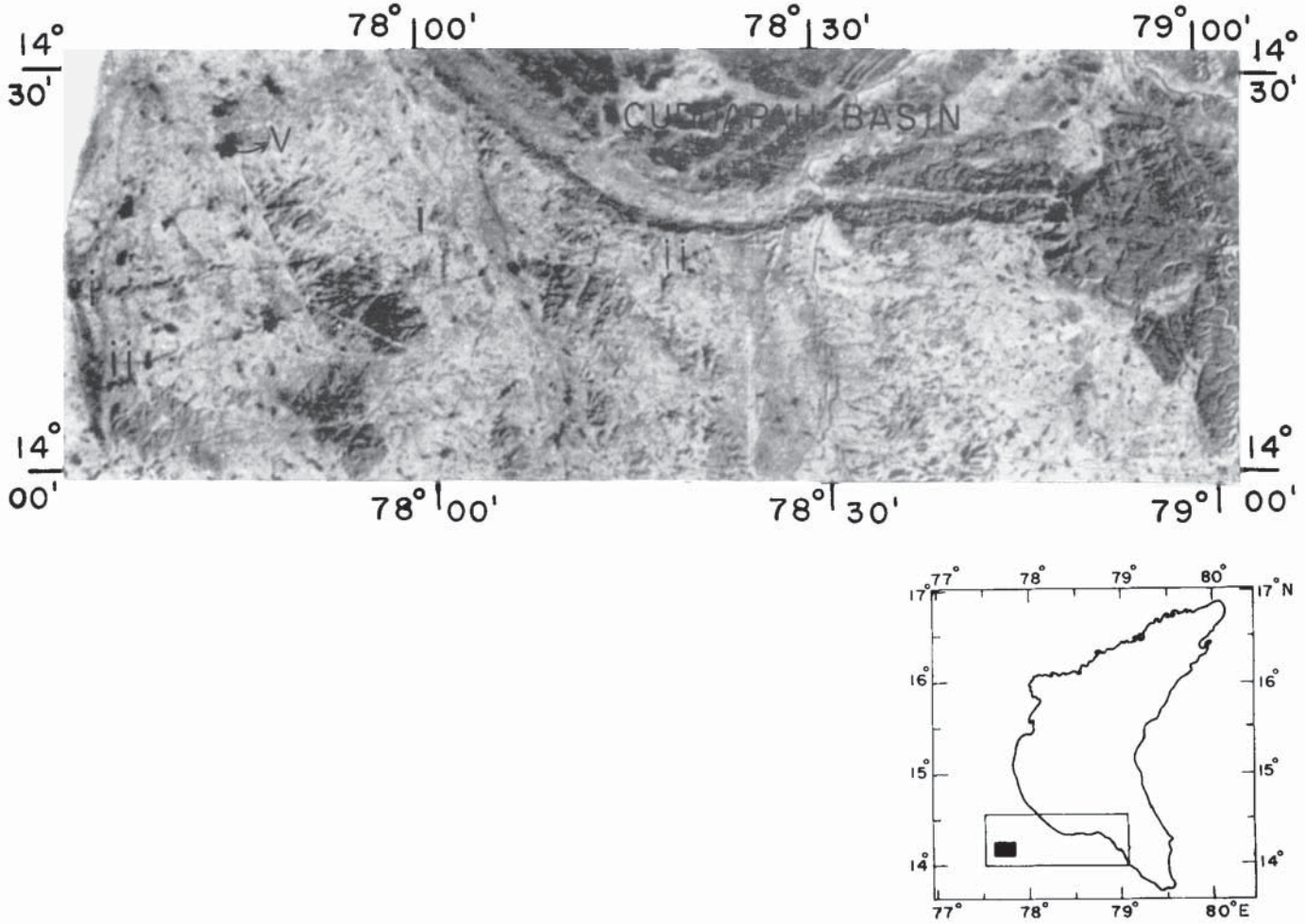
to batch demagnetization at peak fields of 15, 20 and 40 mT for all the specimens. Statistical parameters were computed giving unit weight to each sample as per Fisher (1953). Results showing minimum dispersion are reported in Table 2.

Results and discussions

Though NRM shows fairly consistent directions the grouping improved with cleaning. From vector migration and intensity decay curves it is evident that a strong secondary component had been overprinted on the primary. Vector subtraction analysis has indicated random secondary magnetization. Since this has been successfully removed by magnetic cleaning, the secondary component could be either due to isothermal remanent magnetization (IRM) or the lower coercivity could simply be attributed to the multidomain magnetic fraction.

There is a relationship between the strikes and the RMD (Table 2). Dykes i and ii show similar RMD and strike, and so do dykes iii and iv. Dyke v (striking NW–SE) possesses a stable RMD which is different from the above two groups. Therefore, these five dykes show three distinct groups of RMDs and strikes. This could probably be due to the tensile forces active during the respective epochs developing unidirectional fissures which were subsequently filled up by doleritic material. Considering that the direction of magnetization depends on the orientation of the field at the time of consolidation of the igneous rocks, it can be said that different directions of magnetization suggest different episodes of igneous activity. Therefore, from these results, it is evident that there could have been at least three episodes of igneous activity in the area.

Baked samples collected on dyke iii, near the contact with dyke ii, possess a stable RMD similar to that of dyke ii (Table 2). Therefore palaeomagnetic (baked contact test) and geological investigations indicate dyke ii to be younger than dyke iii. Previous literature (Balakrishna, Rao & Venkatanarayana 1979; Bhattacharji 1981) and the present investigation suggest dyke ii to be older than the Cuddapah basin which means that these two sets of dykes (NE–SW and ENE–WSW or E–W trending) are older than the Cuddapahs. Dykes lying to the south of the basin (Fig. 1) also possess a similar direction of magnetization (Anjanappa 1972) to that of dykes iii and iv. Remanie directions were estimated from total field magnetic anomalies over the dykes beneath the Cuddapah basin (Balakrishna & Paul 1979). This magnetization direction is similar to that of dyke v of the present study. This



[facing page 32]

Plate 1. *Earth Resources Technology Satellite* (*ERTS*) imagery showing the location of the sampled dykes with respect to the Cuddapah basin. The dark portion to the top right is a part of the Cuddapah basin and the rest of the area is the unclassified Archaean crystalline rocks. The prominent dykes in the Archaeans can be seen as dark narrow strips in this imagery. Dykes (i), (ii) and (v) have been marked in the photograph. The other two dykes (iii) and (iv) are not identifiable in this figure as the scale is 1 : 1 000 000. Dykes are numbered as in the text. In the key map the darkened portion shows the area covered by Fig. 2.

Table 3. Palaeomagnetic data of dolerite dykes in the Indian Peninsular Shield.

Group	*no.	Dykes	α_{95}	D_m	I_m	λ_p	L_p	λ_m	Source
(I)	9	Tiptur dyke	13	287	−21	14 S‡	152 E	10 N	Bhalla, Asha Hansraj & Prasad Rao (1980b)
	11	Hansur dyke	33	97	−14	8 S	172 E	6 S	Bhalla, Asha Hansraj & Prasad Rao (1980a)
	9	Tirupati dyke	4	278	−22	5 S‡	156 E	9 N	Anjanappa (1972)
	3	Karimnagar dyke	9	67	+12	23 N	169 E	10 N	Rao & Radhakrishna Murthy (1980)
	6	Anantapur (dyke iii)	7	64	−7	23 N	177 E	1 N	Present study
	6	Anantapur (dyke iv)	6	57	−8	31 N	182 E	4 S	Present study
	13	Coimbatore dyke	5	38	−34	42 N	204 E	10 S	NGRI unpublished data
	4	Hyderabad dyke	5	44	−3	43 N	173 E	11 N	Verma, Pulliah & Hasnain (1968)
	5	Hyderabad dyke	7	32	−16	49 N	205 E	5 S	Kumar & Rao (1979)
	10	Kolar dykes	2	37	−18	47 N	197 E	2 S	Reddy & Prasad (1979)
	1	Gwalior Traps†	18	70	+3	19 N	176 E	1 N	Athavale *et al.* (1963)
(II)	2	Singhbum dyke	6	50	+84	30 S‡	69 E	76 N	Verma & Prasad (1974)
	3	Karimnagar dyke	13	40	−63	17 S‡	50 E	56 N	Rao & Radhakrishna Murthy (1980)
	6	Anantapur (dyke i)	7	57	−69	7 S	47 E	56 N	Present study
	6	Anantapur (dyke ii)	6	71	−72	3 N	47 E	51 N	Present study
	12	Sargur dyke	24	302	+79	23 N	57 E	70 N	Bhalla *et al.* (1980b)
	7	Bangarpet dyke	7	129	−81	24 N	63 E	75 N	Bhalla *et al.* (1980b)
	1	Gwalior Traps†	7	3	−74	5 S‡	77 E	64 N	Klootwijk (1974)
(III)	8	Tirupati dyke	3	349	−17	62 N	75 W	9 N	Anjanappa (1972)
	2	Singhbum dyke	7	330	−9	50 N	53 W	5 N	Verma & Prasad (1974)
	10	Kolar dyke	3	324	−9	50 N	36 W	5 S	Reddy & Prasad (1979)
	10	Kolar dyke	4	318	−6	45 N	30 W	2 N	Reddy & Prasad (1979)
	11	Mysore dyke	7	329	−48	39 N	51 W	23 N	Hasnain & Qureshy (1971)
	5	Anantapur (dyke v)	14	320	−34	38 N	68 W	19 N	Present study

*Numbers given as in Fig. 6; † not dykes; ‡ antipole.
λ_m = ancient latitude of Nagpur (centrally located place in India).
Notations used as in Table 2.

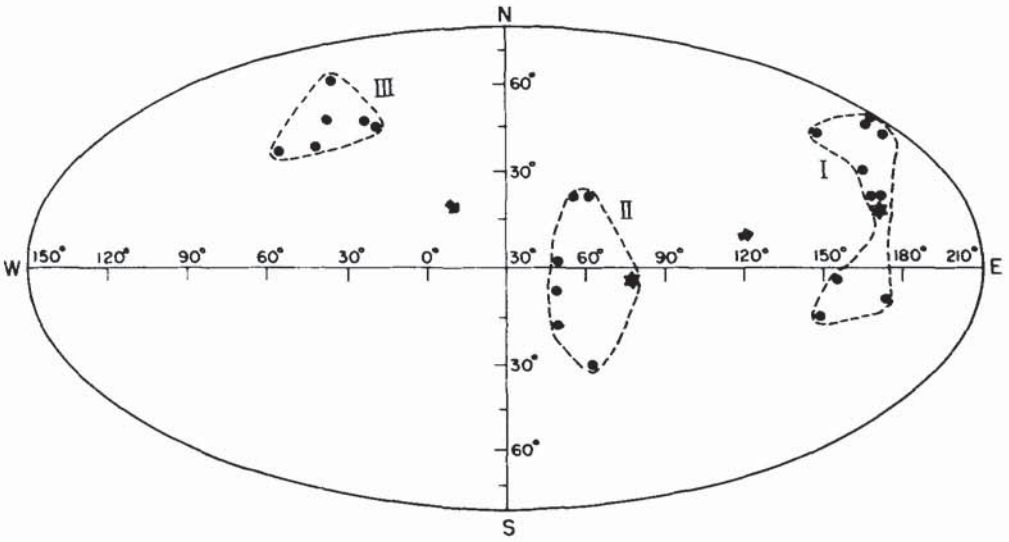


Figure 5. Molleweide's projection showing the Virtual Geomagnetic Pole positions of the data listed in Table 3. These VGPs fall in to three distinct groups indicated by I, II, III from younger to older ages. The groups are numbered as in Table 3. Stars indicate VGPs of Gwalior Traps.

means that most of the dykes in the vicinity of the basin could have been emplaced during the same periods of igneous activity as that of the three phases of dyke intrusions reported in the present investigations.

To establish further the relationship between the dykes and the adjoining Cuddapah basin, palaeomagnetic directions of various Precambrian dykes in the Indian peninsular shield have been synthesized in Table 3, plotted in Fig. 5 and their locations given in Fig. 6. From Table 3 it is evident that the magnetic directions of dykes from different parts of the shield fall into three groups. Magnetic directions and pole positions of group I (Table 3) dykes are similar to those of upper Gwalior traps (Athavale, Radhakrishna Murthy & Sahasrabudhe 1963; Klootwijk 1974) which had been dated by Crawford & Compston (1970) at 1830 ± 200 Myr. These traps may date this group. Klootwijk (1974) has also reported similar RMD for lower Gwalior traps (1830 ± 200 Myr) as that of the group II dykes. None of the group III dykes is dated but Anjanappa (1972) has speculated that Tirupathi dykes with similar directions of magnetization to those of the group III dykes, could have intruded at the closing stages of the Archaean period. Hasnain & Qureshy (1971), however, attribute an age contemporaneous to that of the Deccan traps. Hence no appropriate age can be assigned to this group of dykes. The similarity of the RMD of the Gwalior traps (1830 ± 200 Myr) with the dykes (four of them) of the present study suggests their age to be contemporaneous with them. If this is so then these dykes should be older than the Cuddapahs whose upper age limit is not more than 1700 Myr on the basis of lithologic considerations (Murthy 1979b).

From these studies it is evident that most of the dykes adjoining the Cuddapah basin on the west and south seem to be older than it. If this is true then these dense basic intrusives (thermal events) could have been responsible for the increase in the density of the lower crust which could have triggered off the initiation of this basin (Beloussov 1960, Sheridan 1969 and Haxby, Turcotte & Bird 1976). The thermal activity in the basin is perhaps responsible for the ultimate development of the basin (Bhattacharji 1981).

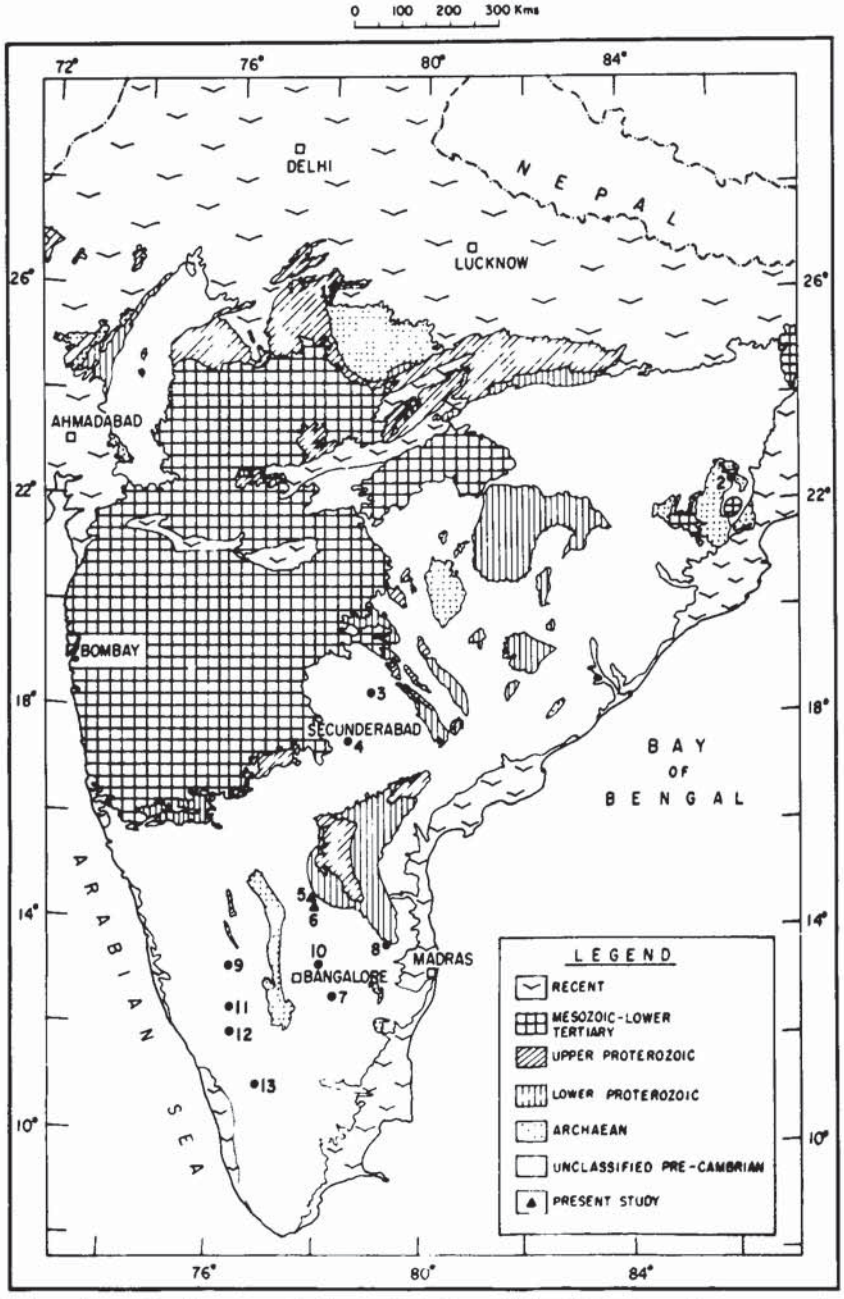


Figure 6. Locations of palaeomagnetically studied Precambrian dykes. Locations are numbered as in Table 3.

Conclusions

There could have been at least three episodes of igneous activity in the area. Besides, dykes with similar directions of magnetization are parallel to each other running in the same general strike direction. This suggests that the tensile forces active during these epochs had developed unidirectional fissures for a single period. Hence three different strike directions

have resulted for the three different epochs. These studies also suggest that the igneous intrusions about and under the Cuddapah sediments are older than them. Therefore, this can be attributed to be the cause for the thermal expansion which initiated the basin formation.

Synthesized palaeomagnetic data of dykes on the Indian shield in general suggest at least three separate periods of dyke emplacements.

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